

Question Paper

Exam Date & Time: 02-Jan-2026 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025/JAN 2026
SUBJECT: MAT 1103/MAT_1103 COMPUTATIONAL MATHEMATICS - I

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Test for consistency and solve the following system of linear equations by Gauss elimination method. (4)

$$2x_1 + x_2 + 2x_3 + x_4 = 6$$

$$6x_1 - 6x_2 + 6x_3 + 12x_4 = 36$$

$$4x_1 + 3x_2 + 3x_3 - 3x_4 = -1$$

$$2x_1 + 2x_2 - x_3 + x_4 = 10$$

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 1B) Find all the eigenvalues and the eigenvector corresponding to the smallest eigenvalue of the matrix. (3)

$$A = \begin{pmatrix} 2 & -1 & 1 \\ 0 & 1 & 6 \\ 0 & 3 & 4 \end{pmatrix}$$

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Use Gauss Seidel method to solve the system of linear equations: (3)

$$3x + y + z = 9.3$$

$$x - 2y + 10z = 30.6$$

$$2x + 5y - z = 10.5$$

Perform 3 iterations and correct the solution up to 4 decimals by taking the initial approximation as.

$$x^{(0)} = 1, y^{(0)} = 2, z^{(0)} = 3.$$

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) Apply Gram Schmidt orthogonalization process to construct an orthonormal basis of from the following linearly independent set (4)

$$S = \{a_1 = (-2, 1, 2), a_2 = (-7, 2, 1), a_3 = (-11, -11, -1)\}.$$

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 2B) (3)

Find the null space of the matrix $A = \begin{pmatrix} 1 & 1 & 3 & 3 \\ 0 & 2 & 2 & 4 \\ 1 & 0 & 2 & 1 \end{pmatrix}$ and hence find its basis and nullity.

$$\begin{pmatrix} 1 & 0 & 2 & 1 \\ 1 & 1 & 3 & 3 \end{pmatrix}$$

Course Outcome: "CO2" / Cognitive Level: "Application"

2C) (3)

Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation such that

$$T(x, y, z) = (x - 3y - 2z, y - 4z, z).$$

Then prove that T is an invertible linear operator on $\mathbb{R}^3$. Also find $T^{-1}(-7, 4, -1)$.

Course Outcome: "CO2" / Cognitive Level: "Application"

3A) (4)

$$\text{Solve } (D^3 - D^2 - 6D)y = e^{3x} + x^2 + 1$$

Course Outcome: "CO3" / Cognitive Level: "Analysis"

3B) (3)

Use method of variation of parameter to solve

$$(D^2 - 3D + 2)y = \sin(e^{-x})$$

Course Outcome: "CO3" / Cognitive Level: "Application"

3C) (3)

Trace the parametric curve $x = 2(t + \sin t), y = 2(1 + \cos t)$ with suitable

justifications.

Course Outcome: "CO3" / Cognitive Level: "Application"

4A) (4)

Diagonalize the matrix $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$ and hence compute A^5 .

Course Outcome: "CO1" / Cognitive Level: "Analysis"

4B) (3)

Use Newton Raphson method to find the approximate real root of

$$f(x) = \sin x - \frac{x+1}{x-1}$$

with initial approximate root $x_0 = -0.4$. Carry out 4 iterations correct to 4 decimal places.

Course Outcome: "CO4" / Cognitive Level: "Application"

4C) (3)

Use Modified Euler's method to find approximate value of $y(1.5)$, given that

$$\frac{dy}{dx} = x^2 e^{(x-y)} \text{ with } y(1) = 0.2. \text{ Take } h = 0.25 \text{ and carryout 3 iterations}$$

in each step, correct to 4 decimal places.

Course Outcome: "CO4" / Cognitive Level: "Application"

5A) (4)

Use Newton's divided difference formula to determine as a polynomial in for the following data.

x	3	6	7	9	10
$f(x)$	168	147	120	72	63

Course Outcome: "CO5" / Cognitive Level: "Analysis"

5B) Use finite difference method to find the value of "a" and "b" from the following table: (3)

x	0	5	10	15	20	25	30
y	1	3	a	73	225	b	1153

Course Outcome: "CO5" / Cognitive Level: "Application"

5C) The following table gives the velocity of a particle at time t . (3)

t (sec)	0	2	4	6	8	10	12
v (m/sec)	4	6	16	34	60	94	136

Use appropriate numerical method to find the distance moved by the particle in 12 sec.

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	MAT_1103:COMPUTATIONAL MATHEMATICS - I

#	Course Outcome	Description	Marks
1	1	CO 1	14
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	6
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	MAT_1103:COMPUTATIONAL MATHEMATICS - I

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	10	30(60%)
4	Analysis	5	20(40%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)